

Compact hulls and support-face selection

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Scope

Compact hulls and support faces provide successive finite refinements and a common generator. The sets need not themselves be convex; nonemptiness and the stated finite-dimensional hypotheses remain in the individual Cards.

4 direct Cards + 0 reused prerequisites = 4 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: [A common generator selected from equal convex hulls](#)
- Key definition: [Successive maximum refinement by an ordered list of functionals](#)

[Read this route with prerequisites](#)

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

0 1

Level 0 (3 Cards)

Level 0

Successive maximum refinement by an ordered list of functionals

Defines a nonempty compact set of survivors after maximizing finitely many functionals in order.

`MathlibAnnex.lexicographicRefine`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

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Level 0

A support face is the convex hull of the maximizing generators

Identifies the points of a convex hull on a supporting level by discarding generators with zero weight.

`MathlibAnnex.convexHull_maxSlice_eq_supportFace`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: [A common generator selected from equal convex hulls](#)

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Level 0

The convex hull of a compact set in a finite real coordinate space

Realizes the ordinary convex hull as a finite union of compact images of bounded-length convex combinations.

`MathlibAnnex.isCompact_convexHull_pi`

Immediate Card prerequisites: None in this selected scope

Used by in this scope: None in this selected scope

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Level 1 (1 Card)

Level 1

A common generator selected from equal convex hulls

Uses a first maximizing slice and a separating sequence to obtain an original common point carrying a prescribed property.

`MathlibAnnex.exists_common_of_convexHull_eq`

Immediate Card prerequisites: [Successive maximum refinement by an ordered list of functionals](#) · [A support face is the convex hull of the maximizing generators](#)

Used by in this scope: None in this selected scope

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